

E12 Multiple Reactions in a Non-isothermal PFR Python solution

Thursday, March 14, 2024

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The gas phase elementary reactions $2A \rightleftharpoons B$ and $2B + A \rightarrow C$ are carried out adiabatically in a PFR with no pressure drop. Plot the temperature and concentrations along the volume of the PFR.

Info:

$k_{1A} = 50 \text{ M}^{-1} \text{ min}^{-1}$ at 350 K with $E = 8 \text{ kJ/mol}$

$k_{2C} = 4000 \text{ M}^{-2} \text{ min}^{-1}$ at 310 K with $E = 4 \text{ kJ/mol}$

$K_{C,A} = 10 \text{ M}^{-1}$ at 315 K

$\Delta H_{1A}^{rxn} = -20 \text{ kJ/mol A}$

$\Delta H_{2B}^{rxn} = +30 \text{ kJ/mol B} \leftarrow \text{convert to C: } \left(30 \frac{\text{kJ}}{\text{mol B}}\right) \left(\frac{2 \text{ mol B}}{\text{mol C}}\right) = 60 \frac{\text{kJ}}{\text{mol C}}$

$V = 1 \text{ L}$

$T(0) = 150^\circ\text{C}$

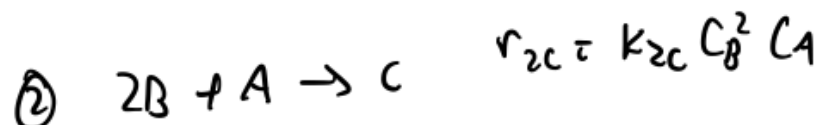
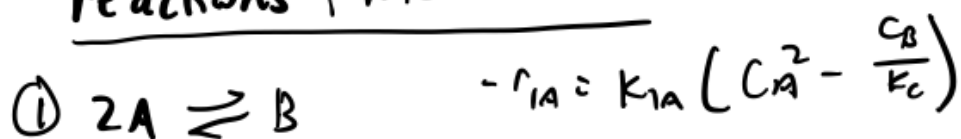
$CA0 = 0.1 \text{ M}$

$FA0 = 1 \text{ mol/min}$

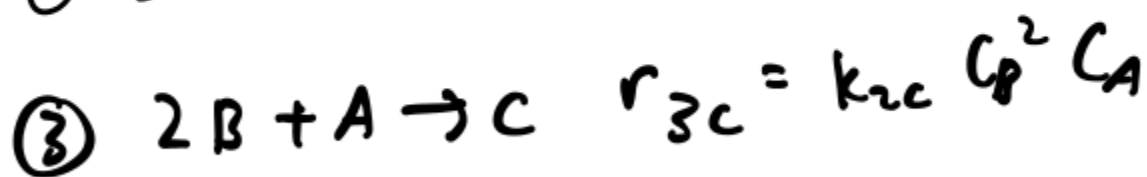
$CPA = CPB = 0.09 \text{ kJ/mol}\cdot\text{K}$

$CPC = 0.18 \text{ kJ/mol}\cdot\text{K}$

reactions & rate laws:



rewrite reactions for stoichiometry matrix



Stoichiometry matrix

$$v = \begin{bmatrix} -1 & \frac{1}{2} & 0 \\ 1 & -\frac{1}{2} & 0 \\ -1 & -2 & 1 \end{bmatrix}$$

rate matrix

$$r = \begin{bmatrix} K_{1A} C_A^2 \\ \frac{K_{1A}}{K_C} C_B \\ K_{2C} C_B^2 C_A \end{bmatrix}$$